# Jin Tao

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## Education

**Technische Universität Berlin** — Berlin, Germany
*M.Sc. in Computational Engineering Science; GPA: 1.3 (1.0 best)* — *Feb. 2025 – Present*

- Thesis and one 6-credit course remaining; to be completed in personal time alongside full-time work.

**Technische Universität Berlin** — Berlin, Germany
*M.Sc. in Vehicle Engineering* — *Sep. 2020 – Feb. 2025*

**Hainan University** — Hainan, China
*B.Sc. in Vehicle Engineering* — *Sep. 2015 – June 2019*

## Experience

**Researcher (full-time)** — Aug. 2026 – Present
*Working Student* — *May 2026 – July 2026*
*Center for Tangible Artificial Intelligence and Digitalization (ZEKI)* — *Berlin, Germany*

- Lead development of a TurtleBot4 robotics platform integrating SLAM, navigation, autonomous exploration, semantic mapping, and an AI agent that translates user requests into robot actions.
- Implement a distributed onboard architecture, running SLAM and navigation on Raspberry Pi and semantic mapping and the agent system on NVIDIA Jetson Nano.
- Develop persistent 3D semantic mapping using an RGB-D camera and 3D LiDAR, with scene-graph representations of objects and their spatial relationships across observations.

**Research Assistant** — Oct. 2025 – Apr. 2026
*Robotics and Biology Laboratory (RBO)* — *Berlin, Germany*

- Designed and built an acoustic gripper from scratch, using acoustic signatures to identify grasped objects.
- Improved the software pipeline for a Franka manipulator by resolving a persistent impedance-controller bug that caused abnormal oscillations at higher impedance settings.
- Restructured the Franka ROS 2 stack and streamlined Git workflows using submodules, establishing clear package responsibilities and boundaries across drivers, controllers, and robot descriptions to improve maintainability and development efficiency.

**Working Student** — Sep. 2024 – Oct. 2025
*Mbition GmbH* — *Berlin, Germany*

- Developed a full-stack software system for monitoring test vehicles and managing vehicle bookings.

**Software Engineer** — Oct. 2022 – Mar. 2024
*TU Berlin Formula Student Team - FaSTTUBe* — *Berlin, Germany*

- Developed a ROS 2 trajectory-tracking and control package using local and global planned trajectories, with pure pursuit and model predictive control (MPC), targeting minimum lap time.
- Developed a steering control system using cascaded current, speed, and position PID loops to achieve fast, stable tracking of commanded steering positions.

**Team Leader** — May 2018 – Nov. 2019
*Hainan University Formula Student Team* — *Hainan, China*

- Designed and built the vehicle's double-wishbone suspension, applying vehicle dynamics principles to determine its geometry.
- Led team organization and coordination, and competed as the endurance-event driver at Formula Student China 2019.

## Achievements and Projects

**TaoRobot: Autonomous Driving Full Stack** | Code & Demo — June 2026 – Present

- Sole developer of an open-source ROS 2 autonomy stack, owning architecture, algorithm implementation, and Gazebo simulation integration.
- Implemented pose-graph SLAM, particle-filter localization, A* path planning with path smoothing, and pure pursuit trajectory tracking.

**Magnetic Joint-State Estimation** Mar. 2026 – Sep. 2026

- Proposed and developed a ball-joint orientation estimation system using two Adafruit magnetic sensors costing less than €7 each and a magnet to estimate yaw, pitch, and roll.
- Evaluated orientation accuracy against an Xsens MTi-630 IMU, achieving comparable accuracy under the tested conditions without long-term drift.
- Preparing a conference paper with the project team.

**Machine Learning for Visuo-Tactile Sensor Fusion** Mar. 2026 – Sep. 2026

- Developed training and evaluation infrastructure for estimating in-hand object pose from RGB images and tactile measurements, using ground-truth pose annotations.
- Implemented and compared ten existing methods, then developed a transformer-based fusion model that matched the best evaluated baseline's pose-estimation accuracy.

**Master's Thesis: Autonomous Shuttle Pedestrian Following** Aug. 2024 – Jan. 2025

- Developed and deployed a pedestrian-following system on a real autonomous shuttle, integrating A* path planning, pure pursuit trajectory tracking, and cascaded PID control.
- Integrated YOLO-based gesture detection to stop, resume driving, and adjust following distance.

**GPU-Accelerated Visual SLAM and Navigation Integration** Mar. 2025 – Sep. 2025

- Deployed GPU-accelerated visual SLAM on a Jetson Xavier-based mobile robot and integrated its outputs with Nav2 by projecting 3D map points into a 2D occupancy grid.
- Enabled GPU acceleration for visual SLAM on NVIDIA Jetson Xavier, reducing CPU utilization by 40% compared with the original CPU-only pipeline.

## Skills

**Programming**: Python, C/C++, MATLAB/Simulink
**Robotics and Simulation**: ROS, ROS 2, Nav2, Gazebo
**Computer Vision and Machine Learning**: YOLO, DINOv3, PyTorch, TensorFlow, TensorRT
**Embedded Systems and Hardware**: Raspberry Pi, NVIDIA Jetson Nano/Xavier, Arduino, ESP32; soldering, PCB design and debugging
**Development Tools**: Git, Docker
**Agentic Development**: Claude Code, Codex; orchestrating coding agents through Git-based workflows, including issues, branches, and pull requests
**AI Tools**: LangChain, LangGraph, Ollama, llama.cpp
**Languages**: English (advanced), German (intermediate), Chinese (native)

## Selected Coursework (Grades 1.0–1.3; 1.0 Best)

**Robotics**: 1.0 | **Applications of Robotics and Autonomous Systems**: 1.0
**Sensor Systems**: 1.3 | **Machine Learning 1**: 1.3 | **Motion Planning**: 1.3
**Numerical Analysis I for the Engineering Sciences**: 1.0